Car Turn Signal

(Indicator)

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1. Problem

The goal of this project was to build one focused embedded task, firstly in the Tinkercad virtual Arduino environment, then with real Arduino equipment. This project is a **car turn signal**. Two buttons turn the left or right signal on and off with an LED and a buzzer. The third button changes the blinking speed and the buzzer tone, and the setting is saved in EEPROM.

2. Design

The program has no `delay()` and no busy-waits. All work is done with interrupts, flags and a state machine in `loop()` (`state`: 0 off, 1 left, 2 right).

- **External interrupts:** left (D2, INT0) and right (D3, INT1) buttons, FALLING edge. The ISR only sets a flag and saves `millis()`, debounce (50 ms) is checked in `loop()`.
- **Timer interrupt:** Timer1, CTC mode, prescaler 64, `OCR1A = 2499`: $f = 16\,\text{MHz}/(64 \cdot 2500) = 100$ Hz, so 10 ms. The ISR increases `ticks` and sets `blinkFlag` every `speed` ticks (30 = 300 ms, 60 = 600 ms).
- **Set button (D4):** read in `loop()` with debounce, switches speed/tone (300 ms/2000 Hz or 600 ms/1000 Hz).
- **EEPROM:** address 0 = version (1), address 1 = speed (30/60), address 2 = tone/10 (100/200). At start the version and the values are checked, wrong values are replaced with defaults. `EEPROM.update()` writes only when the value changed.
- **Feedback:** LEDs, buzzer (`tone()` uses Timer2) and Serial Monitor (9600).

3. Parts lists

Table 1 shows all components used in this project.

Table 1: Parts list

Name	Quantity	Component
U1	1	Arduino Uno R3
S1 S2 S3	3	Pushbutton
D1 D2	2	Yellow LED
R1 R2 R3	3	220 Ω Resistor
PIEZO1	1	Piezo
2	1	Breadboard

4. Circuits

4.1. Wiring circuit and pin connections

Figure 1 shows the circuit built in Tinkercad and Table 2 lists the pin connections.

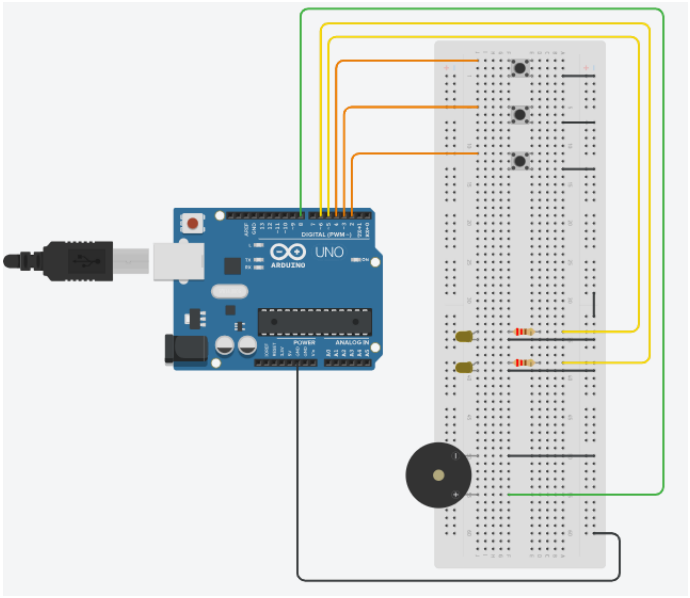


Figure 1: Wiring in Tinkercad

Table 2: Pin connections

Component	**Arduino pin**
Pushbutton (Left turn signal indicator)	D2
Pushbutton (Right turn signal indicator)	D3
Pushbutton (Set different LED speed and buzzer tone)	D4
Yellow LED (through 220 Ω) (Left turn indicator)	D5
Yellow LED (through 220 Ω) (Right turn indicator)	D6
Piezo buzzer (+)	D8
LED cathodes, buzzer (−), pushbuttons opposite side	GND

4.2. Schematic

Figure 2 shows the circuit generated in Tinkercad.

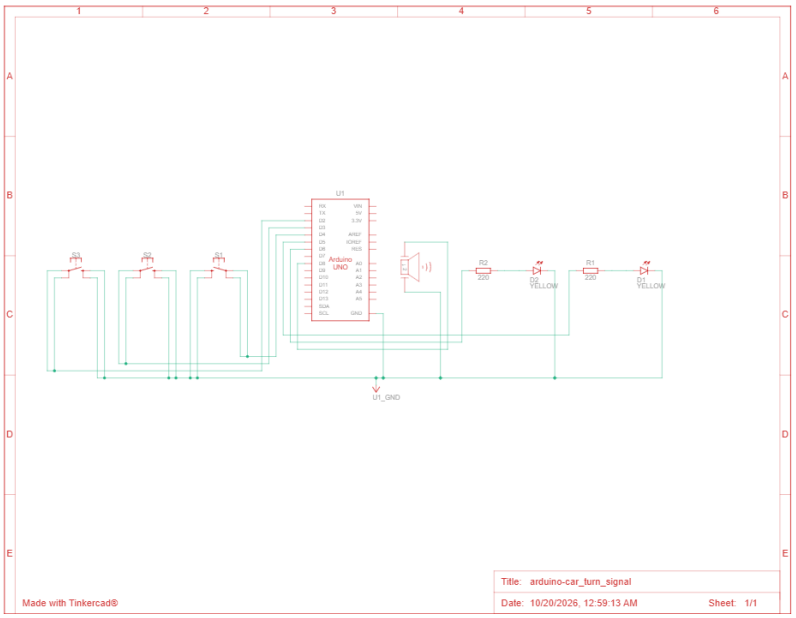


Figure 2: Schematic generated by Tinkercad

4.3. Build with equipment

Figure 3 shows fully built project with real Arduino board implemented with previously Tinkercad environment created wiring circuit (see Figure 1) and pin connections (see Table 2).

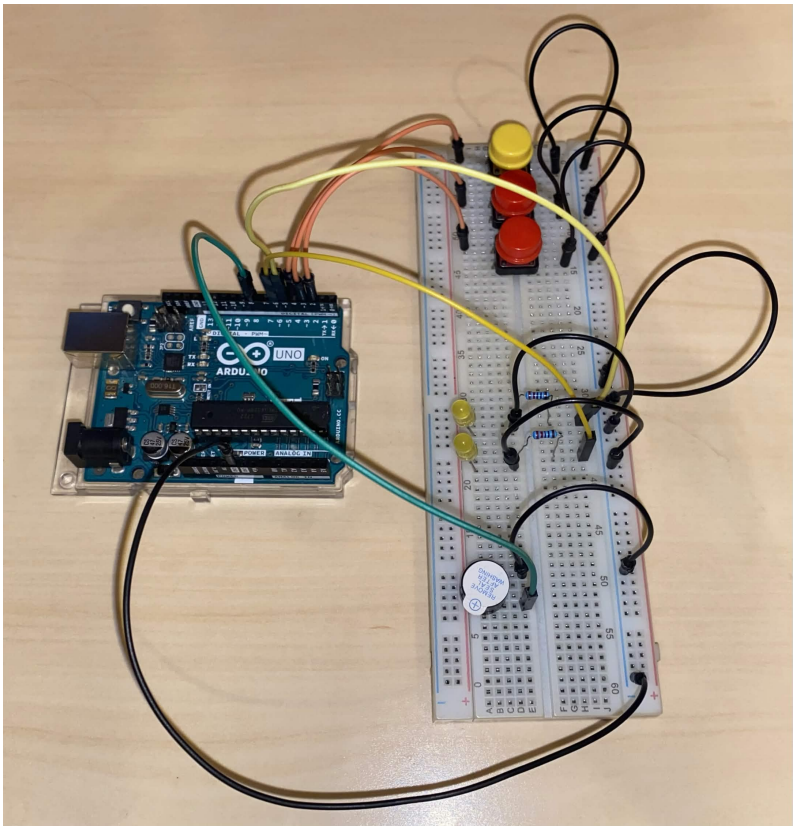


Figure 3: Tinkercad circuit realised with real Arduino

5. Results

5.1. What worked

This list below shows, what worked during this project:

- Left / right signal with LED and buzzer blinking together.
- Speed and tone change with the set button.
- Setting are loaded from EEPROM after the simulation is started again.
- The circuit also works on the real Arduino and breadboard.

5.2. What did not work (and how it was fixed)

Table 3 lists the problems, cause and how they were solved.

Table 3: Problems encountered during development

Problem	**Cause**	**Solution**
The circuit did not work on the real breadboard	Not all components had a good common GND connection, also my mistake, but I forgot that bredboard I use is splitted into 2 parts	Added an extra ground connection, so all parts are fully connected to GND
Button was pressed many times by one press	Button contacts bounce (noise), which makes many falling edges and interrupts	Added debounce: the ISR only saves the time, and `loop()` accepts the press after 50 ms if the pin is still LOW

5.3. Test and accuracy results

Table 4 shows the tests.

Table 4: Test and accuracy results

Test	**Expected**	**Result**
Blink half-period, fast mode	300 ms	299-300 ms
Blink half-period, slow mode	600 ms	599-600 ms
Setting after Arduino restart	last mode	OK, after restart it runs with last mode
Wrong value in EEPROM	default values	OK, it validates and if detects bad input, it returns default values

5.4. Known issues

List below shows, what knows issues were while making this project:

- The set button is read in `loop()`, it is not an interrupt.
- Sometimes button doesn't register as pressed or by one press registers many times (it was encountered during development and fixed, but still persists to happen **SOMETIMES**, not always).

6. Future improvements

To improve this project, next features should be made:

- The set button can be changed to a third interrupt.
- Add a hazard (both sides) mode and more speed levels.
- Fully fix buttons which by one press it registers many times.
- Fix wiring (it is little messy right now).

7. Conclusion

The prototype works as a simple car turn signal. Components such as left/right pushbuttons make the LEDs blink, buzzer beep in real time and set pushbutton change blink speed and buzzer beep frequency.

8. Source code

The full code is available in the repository : `https://github.com/dzenardas/arduino-car_turn_signal`